

Windmill

Immersive Storytelling Experience in Virtual Reality

Tianran, Xia

Virtual Reality Development, Ringling College of Art and Design, Sarasota, United States of America, School of Film and Animation, Rochester Institute of Technology, Rochester, United States of America
troyxia7@gmail.com

KEYWORDS

VR, immersive, interactive, narrative, storytelling, 3D animation, Unreal Engine, Maya, virtual reality

ACM Reference Format:

Tianran, Xia. 2024. Windmill: Immersive Storytelling Experience in Virtual Reality. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Immersive Pavilion (SIGGRAPH Immersive Pavilion '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641521.3664401>

1 INTRODUCTION

Windmill is an interactive narrative experience in virtual reality. It tells the story of an old monochromatic painter named Daniel, who is encouraged by a naughty boy to embrace life by using color. I created this experience to explore the potential of interactive storytelling as a new form of animation. I also experimented with multiple interactions and visual styles to make the experience immersive and convincing for the audience.

1.1 Solve Incompatibility

The most difficult challenge is that conventional storytelling is not compatible with virtual reality. In a traditional film, the director is the storyteller who manipulates everything shown on the screen. However, in virtual reality, a director cannot control the camera because the audience is the one in control of where they look and how they explore the world. Therefore, the essential core of virtual reality is contradictory to the logic of conventional narration, which is controlled by the director. My solution is to make the audience a part of the story in the virtual world and the ones who push the story forward, which helps to set up a connection with the characters. I decided it is most significant to guide the audience to interact with the characters properly, especially for eye contact. Eye contact is an effective tool to break the fourth wall and can convince the audience that the characters in the story are alive. Based on my observation of different play-testers of Windmill, they showed more interest in interacting with the characters instead of considering who they were in the experience. It proved that convincing the audience of their existence in a brand-new world is more attractive than giving them particular identities or roles, at least for beginning virtual reality players. By utilizing these

methods, I successfully created an emotional bond between the audience and the characters. They will interact multiple times before reaching the emotional climax of the story, which allows the audience to realize that they are making changes in this virtual world.

1.2 Optimization

The other challenge is the optimization of this project in Unreal Engine. I created the wheatfield on a gigantic landscape that contained a complicated wind effect. Eventually, I utilized procedural foliage blocking volume to create plants on the wheatfield instead of directly adding dense foliage elements to the huge landscape. Then, I made a couple of wheat materials with different LODs and precise instance numbers to control the overall density of the wheatfield. In addition, because of the difficulty of separating a single object from the post-process volume effect, I separated the colorful boy from the monochromatic world by creating a material parameter collection and adding a saturation parameter to all materials. Therefore, I would only need to change the saturation value in runtime to make the world colorful, and the post-process volume would only be used for lighting and rendering.

1.3 Visual Elements

For the aesthetic style, I created a semi-stylized environment and made it look like a painted world by utilizing different shaders. Before the story begins, the audience will color a black-and-white wheatfield painting in a small and dark room. This process implies to the audience that they will enter the monochromatic world in the painting, which can tremendously enhance their interest and excitement when seeing a bright and colorful painted world at the end of the experience. In addition, the wind plays a significant role in this virtual reality experience. I applied wind effects to the wheatfield, the wires of telephone poles, and the windmill. I also created a lot of windy VFX to bring a feeling of vitality to the audience.

1.4 Conclusion

Virtual reality opens a new realm to explore and experiment with different forms of art, including film and animation. It demonstrates the potential of bringing a new life to the art of animation, which has been greatly explored for the past few decades. It is difficult to define whether Windmill is a film or a game due to its interactive nature, but the process of creating it enlightened me to a new form of art by utilizing this new technology. I will continue to explore interactive storytelling experiences in virtual reality while gradually entering the industry.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH Immersive Pavilion '24, July 27–August 01, 2024, Denver, CO, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0527-4/24/07

<https://doi.org/10.1145/3641521.3664401>

ACKNOWLEDGMENTS

The virtual reality developer would like to thank Ringling College of Art and Design for access to facilities and hardware. This work was

nominated for the Rookie Awards 2023—Rookie of the Year Immersive Media Finalist—and won the Career Opportunities realworld one.